

Interacting Multipoles of Rank 1 and 1:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \mu_\beta \\ -1 \cdot R_z^2 \cdot \delta(\alpha, \beta) \cdot \mu_\alpha \cdot \mu_\beta \end{bmatrix} \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \mu_\beta \\ -1 \cdot R_z^2 \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \mu_\beta \\ -1 \cdot R_z^2 \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \quad (2b)$$

Simplify R:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_z \cdot R_z \cdot \mu_z \cdot \mu_z \\ -1 \cdot R_z^2 \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \quad (3b)$$

Simplify Multipoles:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (4b)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (5a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (5b)$$

Reduce Indices:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (6a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (6b)$$

Add up:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \quad (7)$$

Combining everything:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = \left[-1 \cdot R_z^{-3} \cdot \mu_y \cdot \mu_y \right] \quad (8)$$